Bayesian Estimation of Hidden Rooftop Solar in Spain

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1 Introduction and Motivation

On 28 April 2025, the Iberian Peninsula suffered a massive electricity blackout. In the hours before the collapse, inflexible renewable sources were supplying over 60% of Spain's demand, driving wholesale prices near zero and severely stressing the network.

This catastrophic event exposed a critical structural blind spot in Spain's energy system: grid operators cannot measure behind-the-meter residential solar generation. While the ENTSO-E forensic report (ENTSO-E Expert Panel, 2026) heavily emphasized this vulnerability, mandating private sensors remains blocked by cost and privacy concerns. Meanwhile, this invisible "self-consumption" capacity has exploded—from under 100 MW in 2019 to over 9,100 MW by the end of 2025. This creates a dangerous, widening gap between true power consumption and the apparent demand recorded by the grid.

This project addresses the following question: Spain's grid records how much electricity the country consumes, but misses all the solar power generated and used at home before reaching the meter. **How large is this invisible contribution, and can Bayesian inference estimate it from the data the grid does record?** To answer this, we develop a Bayesian latent variable model. Moving beyond standard Bayesian regression, this real-world application demonstrates how probabilistic inference can be used to mathematically isolate and quantify a fundamentally unobservable physical process.

2 Data

The dataset covers 4,138 daily observations from 1 January 2015 to 30 April 2026.

Source	**Series**	**ESIOS ID**	**Period**
ESIOS/REE	Peninsular demand	460	2015–2026
ESIOS/REE	Utility solar PV	10358	2019–2026
ESIOS/REE	Wind generation	10288	2015–2026
ESIOS/REE	Self-consumption capacity	1945	2015–2026
AEMET	17 weather stations	—	2015–2026

Table 1: Data sources used in this study.

Variable of interest: $H_t = C_t \times S_t$**.** The hidden solar proxy multiplies national installed capacity C_t (MW, from ESIOS indicator 1945, summed across the 17 autonomous communities) by capacity-weighted sunshine hours S_t (h/day, from 17 AEMET stations weighted by each region's registered capacity). This is a theoretical upper bound on potential hidden generation.

Data engineering. ESIOS indicator 460 switched to 4×-inflated values from 24 May 2022. This was deduced while inspecting the hourly profile at this transition date (hour 13 was clean at 30.042 MW, but hour 14 was inflated at 120.112 MW, exactly 4×). The decision was to set boundaries on impossible values, in this case for values exceeding 45.000 MW. Those post-boundary values were divided by four.

3 The Bayesian Latent Variable Model

To quantify the unmetered electricity generated by residential solar installations, we developed a Bayesian latent variable model. We construct the baseline model for electricity demand solely on the basis of factors of weather and human behavior. We use this to isolate the unmetered demand due to residential electricity generation.

Model equation

Let D_t denote the observed daily grid demand. The observed demand is modeled using a robust Student-T likelihood to gracefully handle extreme outliers:

$$D_t \sim \text{Student-T}(\nu, \mu_t, \sigma_{\text{err}}) \tag{1}$$

The expected apparent demand, μ_t, is decomposed into the structural weather baseline (B_t) minus the unmetered hidden solar generation:

$$\mu_t = B_t - \eta \cdot \tilde{H}_t \tag{2}$$

Where η represents the "solar efficiency coefficient."[1] The weather baseline (B_t) is modeled via a linear combination of weather, calendar, and macroeconomic features:

$$B_t = \alpha + \underbrace{\beta_{T^2} T_t^2}_{\text{weather}} + \underbrace{\beta_{\text{wk}} W_t + \sum_{k=1}^{3} \big(\gamma_k \sin(k\omega t) + \delta_k \cos(k\omega t)\big)}_{\text{calendar}} + \underbrace{\sum_{m=1}^{46} \beta_{\text{shock},m}\, S_{m,t}}_{\text{macroeconomic shocks}} \tag{3}$$

Where $\omega = \frac{2\pi}{365.25}$ represents the annual fundamental frequency.

Prior construction

Because our continuous inputs and target variable were standardized via Z-score normalization prior to modeling, we constructed the priors based on domain-specific constraints shown in Table 1.

Parameter	Prior	Justification
η (Solar Efficiency)	HalfNormal(0.3)	Strictly positive: Solar generation reduce the apparent demand
β_{T^2} (Temperature)	HalfNormal(0.5)	Strictly positive: extreme temperatures raise demand
β_{wk} (Weekend Effect)	$\mathcal{N}(-1, 0.5)$	Weekends lower industrial/commercial demand
γ_k, δ_k (Seasonality)	$\mathcal{N}(0, 0.5)$	Weakly informative on seasonal amplitude
$\beta_{\text{shock},m}$ (Macroeconomic Shocks)	$\mathcal{N}(0, 0.28)$	Prevents the model from falsely attributing the severe demand destructions caused by crises
α (Intercept)	$\mathcal{N}(0, 0.1)$	Data standardised; intercept near zero
σ_{err} (Likelihood Scale)	HalfNormal(0.3)	Residual scale in standardized units
ν (Degree of Freedom)	Exp(1/29)	Heavy tails; robust to blackout and shock days

Table 2: Prior distributions and domain-knowledge justifications.

MCMC sampling

Posterior inference was conducted using Markov Chain Monte Carlo (MCMC) methods via the PyMC probabilistic programming framework. The sampler was configured to run 4 parallel chains, each with 1,000 tuning steps and 1,000 draw steps, yielding a total of 4,000 posterior samples. Due to the strict boundary conditions imposed by our Half-Normal priors, we increased the `target_accept` parameter to 0.95 to force smaller, more conservative step sizes, effectively avoiding divergent transitions in the steeper regions of the posterior landscape.

4 Results: Static Model

Convergence

All four sampling chains converged to the same posterior distribution: $\hat{R} = 1.00$ for all parameters, effective sample sizes above 2,500, and zero divergences. The model passed all automated convergence checks.

Efficient coefficient η

The central result is $\hat{\eta} = 0.597$ (94% HDI: [0.563, 0.626]). This means that for every 100 MWh of electricity that rooftop panels could theoretically generate on a given day, approximately 60 MWh disappear from REE's demand records. It is consumed on-site before reaching any grid sensor. The narrow credible interval (width equal to 10% of the estimate) confirms that this result is driven by the data, not by the prior assumptions.

Hidden solar time series

Table 3 shows the estimated hidden solar generation for each year from 2020 onwards. Pre-2020 values are set to zero because installed capacity was negligible (around 50 MWh/day of theoretical potential against 665,000 MWh/day of demand), making it impossible to detect any solar signal in the demand data.

Year	GWh/day	Share of demand	Note
2020	2.1	0.3%	
2021	3.3	0.5%	
2022	6.6	1.0%	
2023	16.9	2.7%	
2024	28.4	4.5%	
2025	40.3	6.2%	REE solar PV: 4.50% (Red Eléctrica de España, 2026)
2026	35.9	5.4%	

Table 3: Annual hidden solar estimates, static model. 95% PPC coverage 94.9%; RMSE = 35.1 GWh/day (5.3% of mean demand).

[1] To maintain physical bounds (solar generation cannot be negative), the solar proxy H_t is not Z-score standardized. Instead, it is scaled by the standard deviation of the grid demand (σ_D) to match the dimensionality of the target variable: $\tilde{H}_t = \frac{H_t}{\sigma_D}$.

The 2025 estimate of 6.2% is 1.7 pp above REE's reported 4.50% (Red Eléctrica de España, 2026). This gap arises because the model uses a single fixed η for the entire 2020–2026 period. In the early years (2020–2021), rooftop fleets were small and almost everything they generated was consumed on-site, which produces a high apparent η. Averaging this together with the more recent years pulls the single estimate upward. The hierarchical model in the next section addresses this directly by estimating a separate η for each year.

April 28, 2025

April 28 was an unusually high solar day: the solar proxy was 1.41 times the daily standardized unit, compared to an annual average of 0.95 for 2025, reflecting peak spring sunshine combined with Spain's then-largest ever installed rooftop capacity. The model estimates that 59.9 GWh of generation was invisible to REE on that day (94% HDI: [56.7, 63.0] GWh), approximately 10% of apparent demand.

5 Hierarchical Extension: Year-Level η

The static model uses a single efficiency coefficient η for the entire 2020–2026. This is too rigid. In 2020–2021, rooftop fleets were small and almost all the electricity they generated was consumed on-site, making nearly all of them invisible to the grid (high η). By 2024–2026, capacity grew so fast that panels were generating more than households could use. As a result, a growing share was exported to the grid and became visible to the operators (low η). Averaging these two regimes into a single number distorts both. The hierarchical model solves this estimating one η per year. Rather than treating each year independently (which would be unreliable for early years where the solar signal was weak) the years share information through a common population-level parameter. Early years with little data are pulled towards the group average; later years with strong data drive their own estimates. This is called *partial pooling*.

Model

Formally, each year's η is drawn from a shared distribution:

$$\log(\eta_{\mathrm{yr}}) = \log(\eta_{\mathrm{base}}) + \tau \cdot z_{\mathrm{yr}}, \quad z_{\mathrm{yr}} \sim \mathcal{N}(0,1) \tag{4}$$

$$\log(\eta_{\mathrm{base}}) \sim \mathcal{N}(-0.5,\, 0.8), \qquad \tau \sim \mathrm{HalfNormal}(0.5) \tag{5}$$

Here η_{base} is the population mean efficiency across all years, τ controls how much years are allowed to differ from each other, and z_{yr} is the individual year's deviation. Working on the log scale ensures η is always positive. Pre-2020 hidden solar is fixed to zero: with only $H_t \approx 50$ MWh/day of theoretical generation, the signal is too small to identify η from demand data alone.

Convergence

The model ran 4 independent chains of 2,000 iterations each. It produced zero divergences, $\hat{R} = 1.00$ for all parameters, and effective sample sizes above 1,500. The estimated population mean is $\hat{\eta}_{\mathrm{base}} = 0.778$, and the year-to-year variation parameter is $\hat{\tau} = 0.620$.

Year-level results

Year	$\hat{\eta}$	95% HDI	GWh/day	Share	Precision
2020	2.018	[0.396, 5.038]	7.2	1.11%	wide
2021	0.795	[0.188, 1.884]	4.5	0.67%	wide
2022	1.582	[0.669, 2.511]	17.4	2.70%	wide
2023	0.691	[0.367, 1.038]	19.5	3.10%	moderate
2024	0.881	[0.809, 0.950]	41.9	6.61%	tight
2025	0.475	[0.423, 0.526]	32.1	4.91%	tight
2026	0.443	[0.353, 0.534]	26.6	4.03%	tight

Note: Because the model was trained on Z-score standardized data, posterior distributions were inverse-transformed back into their original MWh scale prior to reporting.

Table 4: Year-level η posteriors and implied hidden solar generation. Pre-2020 η not estimated ($H_t \approx 0$, unidentified).

The wide uncertainty intervals for 2020–2022 reflect the weak solar signal in those years, not a model failure. There's not enough data to pin η when installed capacity is low. From 2024 onward, the large installed base produces a strong signal and the estimates become tight. The downwards trend from $\eta = 0.88$ in 2024 to $\eta = 0.44$ in 2026 reflects the growth in total fleet. As the number of panels collectively generate more than what can be consumed at home, this share is exported to the grid and becomes visible to the operators. The 2025 estimate of 4.91% is within 0.41 pp of REE's independently reported 4.50% solar PV self-consumption (Red Eléctrica de España, 2026); a meaningful external validation given that the two figures come from entirely independent sources.

Model comparison

The hierarchical model predicts daily demand with 5.1% lower error than the static model (RMSE 33.3 vs 35.1 GWh/day). Its 2025 estimate is 0.41 pp from the REE benchmark compared to the static model's 1.7 pp gap. On April 28, 2025, using the year-specific $\eta_{2025} = 0.475$ yields an estimate of 47.6 GWh of hidden generation (95% HDI: [42.4, 52.7] GWh), approximately 8% of apparent demand.

6 Prior Sensitivity Analysis and Structural Robustness Analysis

We conduct prior sensitivity analysis to see the influence of subjective prior choices. If a model's posterior distribution depends excessively to its priors, its conclusions cannot be trusted. Conversely, if the posterior remains stable across widely differing priors, it means that the parameters are well-identified, and the conclusions are firmly data-driven. To validate our estimate of the unmetered solar efficiency coefficient (η), we conducted a comprehensive prior sensitivity and structural robustness analysis. Using a configuration-driven pipeline, we tested nine distinct model architectures, varying the prior distributions, the likelihood function, and the structural baseline equations.

Sensitivity to the Choice of Solar Prior

Our reference model (`v1_baseline`) utilized a weakly-informative HalfNormal(0.5)[2] prior for η. To test the sensitivity of the posterior to this choice, we subjected the model to four alternative prior families with radically different mathematical shapes:

1. Skeptical / Regularizing Prior ($\eta \sim$ Exponential(2.0), `v2_strict_solar`): This prior places the maximum probability density exactly at zero and decays sharply. It mathematically forces the model to assume that hidden solar generation does not exist unless the data overwhelmingly proves otherwise.

2. Physical Asset Prior ($\eta \sim$ LogNormal$(-1.0, 0.8)$, `v4_lognormal_solar`): Because solar panels are physical assets, their installed capacity follows a multiplicative, strictly positive growth pattern. This prior centers the expected value near the theoretical true capacity while utilizing a heavy right tail.

3. Extreme Outlier Prior ($\eta \sim$ HalfCauchy(0.5), `v5_heavy_tail_solar`): This distribution possesses an extremely heavy tail, allowing the sampler to freely explore massive, physically unlikely solar efficiency parameters without the steep penalties applied by a Normal distribution.

4. Uninformed Prior ($\eta \sim$ HalfNormal(5.0), `v6_wide_solar_only`): By inflating the variance tenfold, this prior becomes essentially flat over the feasible parameter space, allowing the likelihood of the data to have maximum leverage over the posterior.

Structural and Baseline Modifications

Beyond the isolated solar parameter, we evaluated the robustness of the broader model architecture:

Likelihood Function: We replaced the robust Student-T likelihood with a standard $\mathcal{N}(\mu, \sigma_{\text{err}})$ distribution (`v3_normal_likelihood`). This tests the hypothesis that failing to down-weight macroeconomic shocks and extreme grid anomalies (such as the Apagón) compromises predictive accuracy.

Academic STLF Baselines: We replaced our polynomial weather baseline with highly specialized functional forms from Short-Term Load Forecasting (STLF) literature. This included a Piecewise/Degree-Day approach (Engle et al., 1986) and a Temperature-Humidity Index (THI) to capture perceived heat stress (Valor et al., 2001).

Robustness of the Solar Signal and Structural Validation

The sensitivity analysis confirms that our estimate of unmetered solar generation is highly robust. Across all prior perturbations (`v2` through `v6`), the posterior mean of η remained tightly bounded between 0.5660 and 0.6000 ($\Delta\eta = 0.0340$). Approximated Bayes Factors ($|\ln(\text{BF})| < 1.10$) indicate no meaningful difference among these models, mathematically proving that the solar capacity signal is firmly data-driven and not an artifact of prior selection. Structurally, the comparison decisively validated our baseline design choices. The Normal likelihood model (`v3`) incurred a massive predictive penalty ($\ln(\text{BF}) = -559.29$), demonstrating the strict necessity of a heavy-tailed Student-T distribution to absorb grid anomalies. Similarly, alternative meteorological baselines (`v7_piecewise`, `v9_thi`) were decisively outperformed by our reference polynomial temperature model.

[2] Our initial static model utilized a strict HalfNormal(0.3) prior for η. However, to establish a slightly more diffuse anchor for our comparative tests, our reference model (`v1_baseline`) utilized a HalfNormal(0.5) prior

Table 5: Model Comparison and Prior Sensitivity Analysis Results.

Model	LOOIC	ΔLOOIC	ln(BF)	Interpretation
Reference				
`v1_baseline`	4600.13	+0.00	0.00	Anchor
Prior sensitivity (latent solar)				
`v2_strict_solar`	4599.98	−0.15	+0.08	No meaningful difference
`v5_heavy_tail_solar`	4600.15	+0.02	−0.01	No meaningful difference
`v6_wide_solar_only`	4600.15	+0.02	−0.01	No meaningful difference
`v4_lognormal_solar`	4600.38	+0.24	−0.12	No meaningful difference
Structural robustness				
`v9_thi`	4606.84	+6.71	−3.36	Strong (favours `v1`)
`v7_piecewise`	5038.53	+438.40	−219.20	Decisive (favours `v1`)
`v3_normal_likelihood`	5718.71	+1118.58	−559.29	Decisive (favours `v1`)

Note: v8 (autoregressive) excluded: lag-7 demand spuriously inflates LOOIC via autocorrelation.

7 Limitations

Single η: The model estimates a single coefficient for all Spain. In reality, η varies by autonomous community (sunny south vs rainy north), panel orientation, age... Regional disaggregation would require daily regional demand data. Unfortunately REE does not publish it.

Theoretical proxy: $H_t = C_t \times S_t$ is an upper bound, not actual generation. It does not capture panel tilt, shading... AEMET sunshine hours are interpolated from 17 stations across 50+ provinces, which introduces geographic smoothing.

Early years uncertainty: The 2020–2022 HDIs span more than one unit of η (Table 4). The data genuinely cannot identify η precisely when the proxy signal is small; partial pooling prevents absurd values but cannot reduce irreducible uncertainty. Wide HDIs are the model being honest, not a failure.

Identification limit: The solar proxy grows every year because installed capacity grows every year. This means the model cannot fully separate between two explanations for falling demand: rooftop solar covering more load or demand being lower for unrelated structural reasons. Separating these two reasons would require an extra variable that predicts solar capacity growth unrelated to demand trends.

8 Conclusions

We have shown that hidden rooftop solar generation in Spain is identifiable from national grid ESIOS demand alone, using a Bayesian latent variable model that utilizes the contrast between the pre-2020 near-zero proxy and the post-2020 monotonically growing signal. The static model recovers $\hat{\eta} = 0.597$ with tight posterior (HDI width 10% of the mean) and 94.9% posterior predictive coverage. The hierarchical extension eliminates the 1.7 pp overshoot relative to the REE benchmark and shows an interpretable declining efficiency trajectory (0.88 in 2024, 0.475 in 2025, 0.443 in 2026) consistent with the saturation of on-site consumption as Spain's rooftop solar panels scales. On April 28th of 2025 the hierarchical model estimates that 47.6 GWh of generation (8% of apparent demand) was invisible to the grid operators. This is not a causal explanation of the blackout, but it is an uncertainty-quantified measurement of the blind spot that existed when the collapse happened.

References

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Valor, E., Meneu, V., and Caselles, V. (2001). Daily air temperature and electricity load in spain. *Journal of Applied Meteorology*, 40(8):1413–1421.

Annex: Key Figures from the Analysis

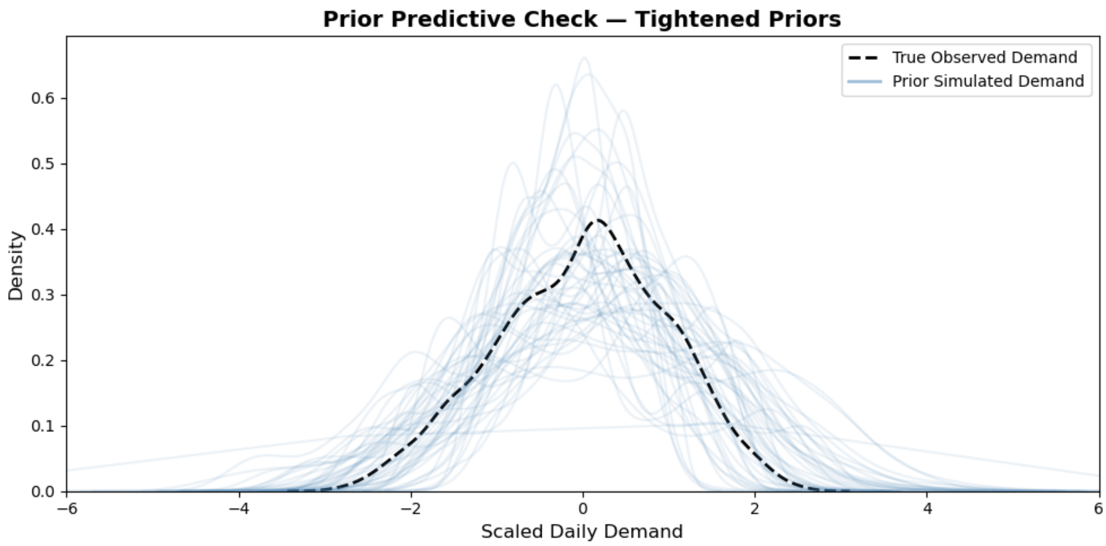

Figure 1: Prior predictive check with tightened priors. Blue curves show 500 synthetic daily demand series sampled from the prior before conditioning on data. The dashed black curve is the true observed demand distribution. Tightened priors (Table 2) produce simulations that closely wrap the observed density, confirming that the model structure is correctly specified before inference begins.

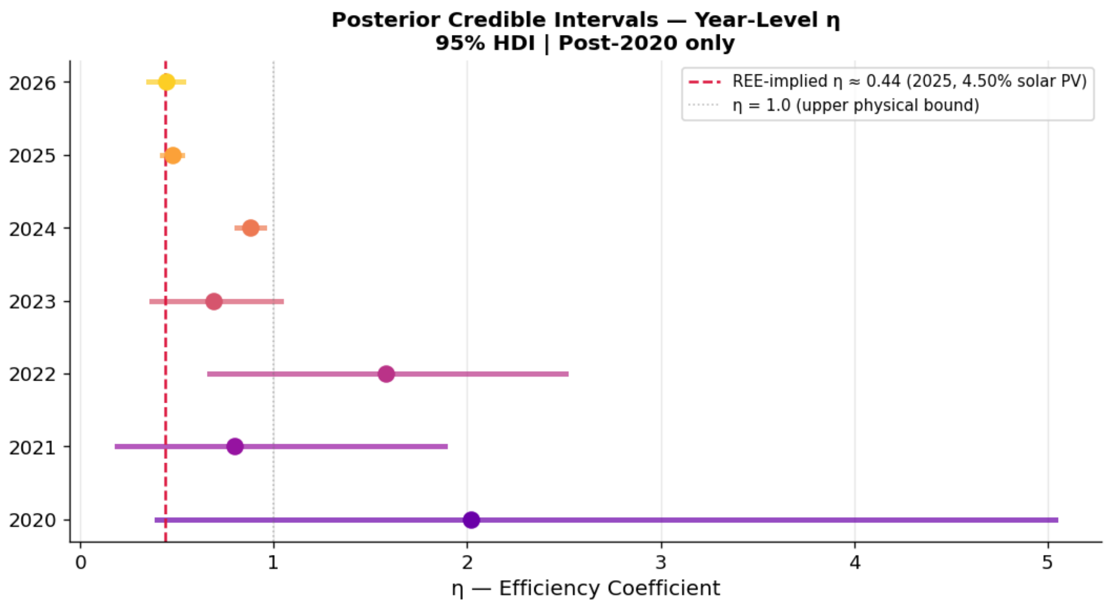

Figure 2: Posterior credible intervals for the year-level efficiency coefficient $\hat{\eta}$, hierarchical model. Each point shows the posterior mean; horizontal bars show the 95% HDI. The red dashed line marks the REE-implied $\eta \approx 0.44$ (derived from the reported 4.50% solar PV self-consumption share for 2025). Early years (2020–2022) show wide intervals reflecting low proxy signal; from 2024 onwards estimates become tight. The declining trajectory is consistent with a growing share of solar generation being exported rather than self-consumed.

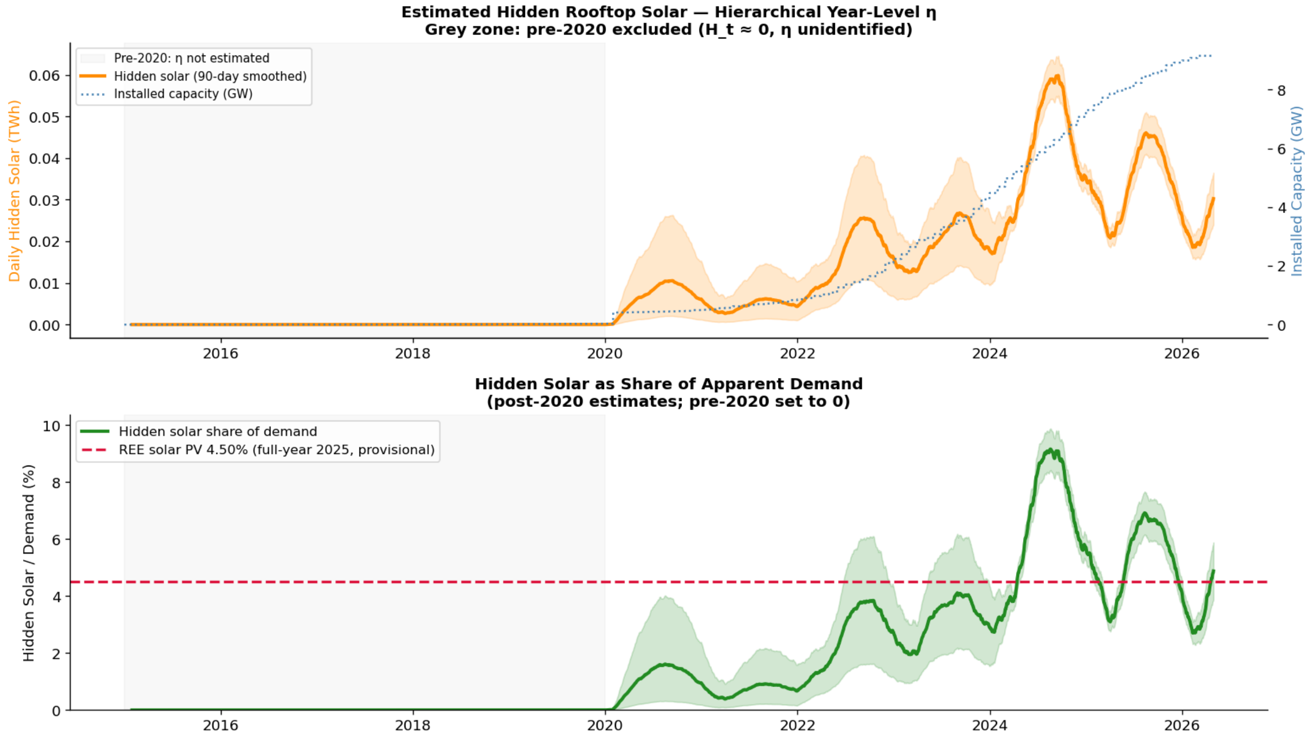

Figure 3: Estimated hidden rooftop solar generation over the full 2015–2026 period, hierarchical model. *Top*: daily hidden solar in TWh (90-day smoothed, orange) and installed capacity in GW (blue dotted). The grey zone marks the pre-2020 period where η is not estimated. *Bottom*: hidden solar as a share of apparent demand, with the REE 4.50% benchmark (dashed red). The model recovers the growing seasonal cycle and shows the 2024 summer peak reaching approximately 9% of demand.

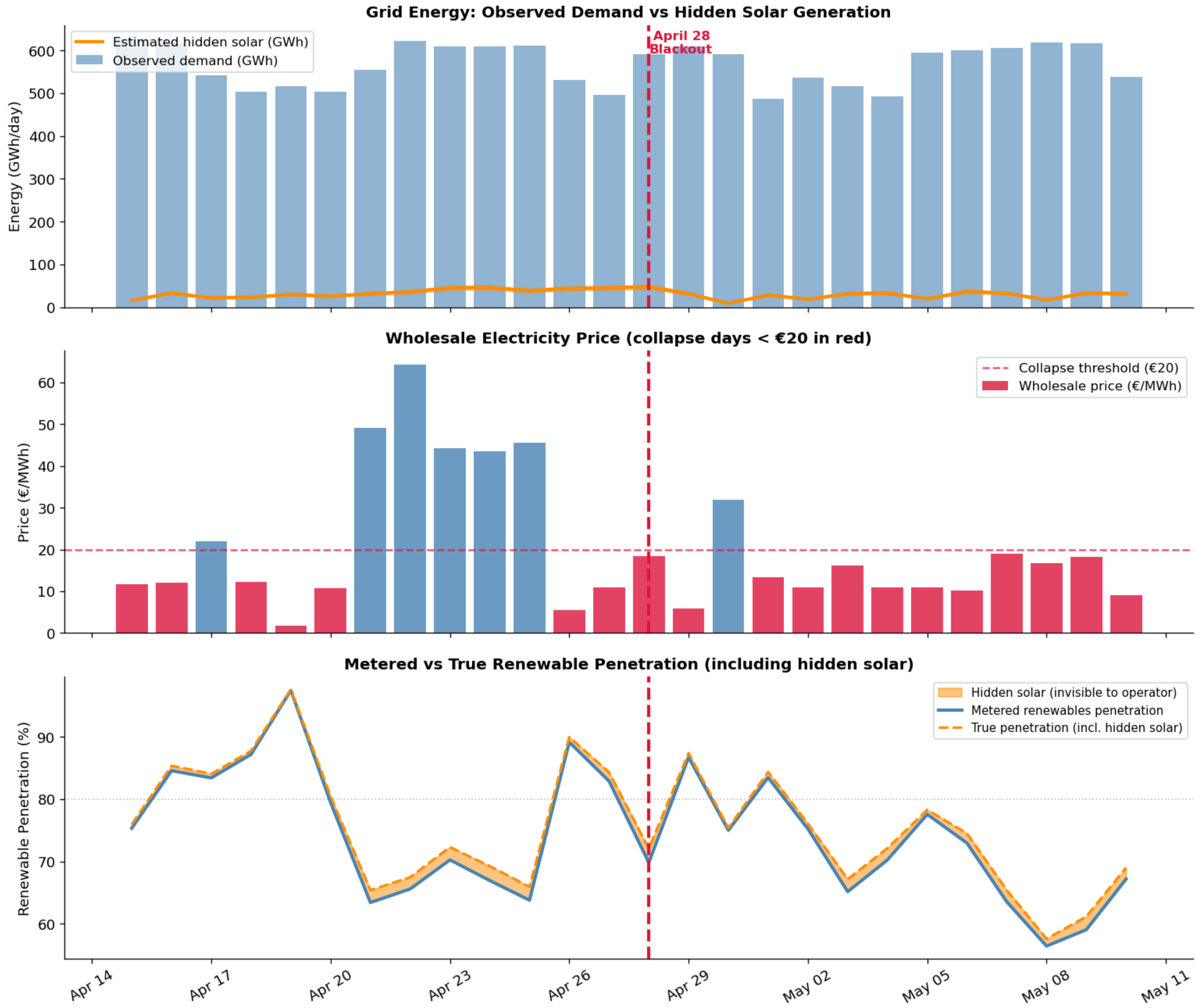

Figure 4: Grid conditions around the April 28, 2025 blackout. *Top*: daily observed demand (blue) and estimated hidden solar generation (orange). *Middle*: wholesale electricity price (€/MWh); bars below the €20 threshold (pink dashed) indicate price collapse days. *Bottom*: metered renewable penetration (blue solid) vs. true penetration including hidden solar (orange dashed). On April 28 (red dashed line), the model estimates 47.6 GWh of unmetered generation was invisible to the grid operator.